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Design and Verification of a Dual-Clock Asynchronous FIFO Gray-Coded Pointer Crossing, Scoreboard Verification and Hardware Validation on PYNQ-Z2

Self Project Report

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M.Tech. Electrical Engineering

(Electronic Systems)

Batch: 2025-27

August 2026

Abstract

This project implements a dual-clock asynchronous FIFO in Verilog-2001, for transferring data between two clock domains that have no fixed timing relationship. Pointers are Gray coded and passed through two-flop synchronisers, so that a pointer sampled by the opposite domain is never a value the counter did not actually hold.

The design was verified with a randomised scoreboard testbench under ModelSim, cross-checked under a second simulator, Vivado XSim, and synthesised for the Zynq XC7Z020 with no inferred latches and no warnings. It was then implemented on a PYNQ-Z2 board, with the write domain clocked from the processing system PLL and the read domain from the board's independent 125 MHz oscillator, where it transferred 286,176,459 words with zero errors while reaching both the full and empty conditions.

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1. Introduction

Most digital systems contain more than one clock. A processor runs at one frequency, an interface at another, and data has to pass between them. Where two clocks have no fixed phase relationship, a signal sampled by the receiving domain may be captured while it is still changing, and the value read can be one that was never actually present.

An asynchronous FIFO is the standard structure used to cross such a boundary. It buffers data between the two domains and reports to each side whether it is safe to write or to read. The difficulty is that the full and empty conditions depend on comparing pointers maintained in different clock domains, so those pointers must themselves cross safely.

Two mechanisms are used together. Gray coding ensures that consecutive pointer values differ in exactly one bit, so a pointer sampled mid-transition returns either the old value or the new one. Two-flop synchronisers give a flip-flop that has been driven into a metastable state a full clock period to settle before anything downstream uses its output. Gray coding does not prevent metastability, and a synchroniser does not prevent multi-bit skew, so both are required.

1.1 Objective

To design a parameterised dual-clock asynchronous FIFO in Verilog, verify it in simulation with a scoreboard and with assertions, confirm through synthesis that the implementation matches the intent, and demonstrate it running on an FPGA with two independent clock sources.

2. Design

2.1 Architecture

The design consists of two files. `sync_2ff.v` is a parameterised two-flop synchroniser, kept as a separate module so that its use is explicit, and `async_fifo.v` contains the memory, the two pointer counters, the Gray conversion and the flag logic. The structure follows the standard arrangement described by Cummings [1].

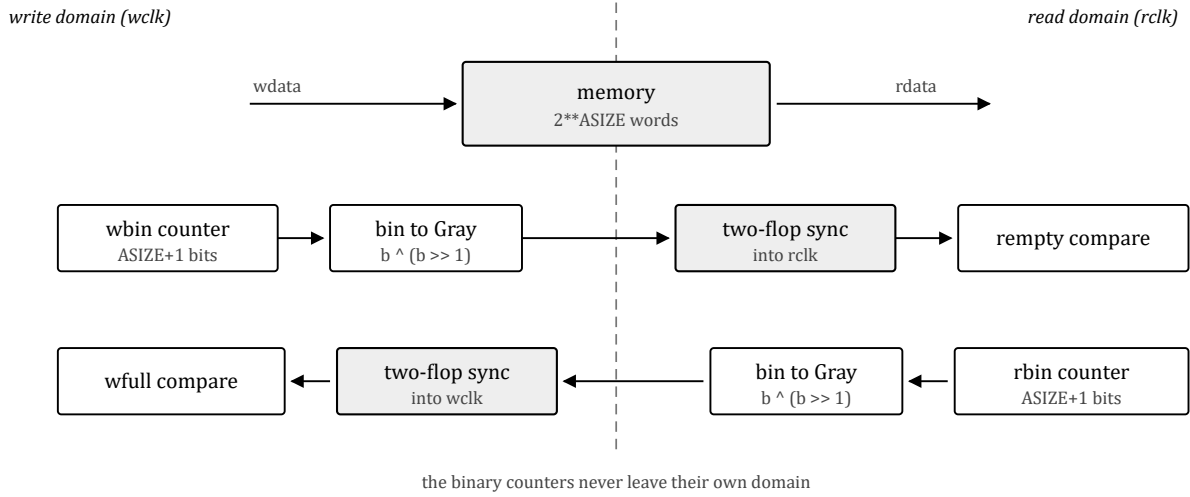


Figure 1: Block diagram of the dual-clock FIFO. Each pointer crosses once, in one direction, as a Gray-coded copy passing through a two-flop synchroniser. The binary counters themselves never cross the boundary.

2.2 Pointers and Flags

Pointers are ASIZE+1 bits wide while the memory address uses only the low ASIZE bits. The extra most-significant bit acts as a wrap flag, and it is the only thing that distinguishes the full condition from the empty condition, since the addresses are equal in both cases. Conversion to Gray code is a single exclusive-or with a right shift:

$$\text{gray} = \text{bin} \wedge (\text{bin} \gg 1) \quad (1)$$

The FIFO is empty when the read pointer has caught the write pointer with no wrap between them, so every bit of the two Gray pointers agrees:

$$\text{rempty} = (\text{rgray_next} == \text{rq2_wptr}) \quad (2)$$

The FIFO is full when the write pointer has wrapped once more than the read pointer and caught it. In Gray code this requires the low bits to agree while the top *two* bits are inverted:

$$\text{wfull} = (\text{wgray_next} == \{\sim \text{wq2_rptr}[\text{ASIZE}], \sim \text{wq2_rptr}[\text{ASIZE}-1], \text{wq2_rptr}[\text{ASIZE}-2:0]\}) \quad (3)$$

Two bits are inverted, not one. In Gray code both the top bit and the bit below it change when the counter passes the halfway point, so inverting only the most-significant bit would detect that halfway crossing rather than the wrap.

Both flags are computed from the *next* pointer value rather than the registered one, so they assert on the same clock edge on which the FIFO actually becomes full or empty. Because each domain sees a copy of the opposite pointer that is two clocks old, the FIFO may report itself full slightly after space has been freed, or empty slightly after a word has been written. The error is always in the safe direction, which is why the design cannot overflow or underflow.

2.3 Module Interface

Table 1: FIFO module interface

Signal	Dir	Domain	Function
wclk, wrst_n	in	write	Write clock and active-low asynchronous reset
winc	in	write	Write request, ignored when wfull is asserted
wdata	in	write	Write data, DSIZE bits
wfull	out	write	FIFO cannot accept a further word
rclk, rrst_n	in	read	Read clock and active-low asynchronous reset
rinc	in	read	Read request, ignored when rempty is asserted
rdata	out	read	Read data, valid whenever rempty is low
rempty	out	read	FIFO holds no unread word

The FIFO is parameterised in data width (DSIZE) and address width (ASIZE). All results reported here use an eight-bit datapath and a depth of sixteen words.

3. Verification

3.1 Simulation Testbench

A single directed test establishes little about a clock-crossing design, so the testbench stalls both interfaces pseudo-randomly and runs the two clocks at a ratio the design was not written for. Two phases execute in one simulation: a fast writer against a slow reader, at 7 ns and 11 ns, and then the reverse. Periods such as 10 ns and 20 ns were avoided because their edges align periodically, which would hide the behaviour the test is meant to exercise.

A reference queue is pushed on the write side, and popped and compared on the read side. Each phase ends by stopping the writes and reading until the FIFO is empty, so that every word written is compared on the way out. Without this drain a few hundred words remain in the FIFO at the end of a phase and are never checked.

3.2 Checks and Coverage

The whole project is Verilog-2001, which has neither assertions nor covergroups, so the checks are written procedurally inside the testbench and the coverage points are counted by hand. Table 2 lists what is verified on every transfer. The same testbench is then run under a second simulator, Vivado XSim, as a cross-check: a design that passes on one tool and fails on another usually has a race the first happened to resolve in its favour.

Table 2: Checks applied during every run

Check	What it verifies
Data	Every word read is compared against the reference queue
Order	The reference queue is FIFO, so any reordering appears as a mismatch
Overflow	Occupancy, taken from the design's own pointers, never exceeds the depth
Underflow	No read is accepted before a matching word has been written
Drain	Every word written during a phase is read back before that phase ends

Four coverage counters record whether the interesting conditions were actually reached: cycles spent full, cycles spent empty, back-to-back writes, and simultaneous read and write. A test that passes without ever filling or emptying the FIFO has demonstrated very little, so the testbench reports these alongside every result and declares a run inconclusive, rather than passed, if any of them is zero.

3.3 Fault Injection

To confirm that the verification is capable of detecting a defect, the full comparison of Equation (3) was deliberately altered to invert only the most-significant bit, and the whole flow was re-run. The scoreboard and the occupancy check reported failures within 500 ns of simulation time, the occupancy check printing *OVERFLOW: occupancy 17, depth is 16*. The correct comparison was then restored.

4. Results

4.1 Simulation

Table 3: Simulation results under both simulators

Metric	ModelSim	XSim
Words written and compared	10,017	10,017
Data mismatches	0	0
Overflows / underflows	0 / 0	0 / 0
Undrained words	0	0
Cycles at full / at empty	12,579 / 12,611	12,609 / 12,592
Back-to-back writes	2,991	2,996

Both boundary conditions were reached in quantity, which is what makes the zero mismatch count meaningful. The two simulators agree exactly on the word count; the coverage counters differ by a few, because they are sampled on clock edges whose relative alignment the two tools resolve slightly

differently.

4.2 Synthesis

Table 4: Out-of-context synthesis results (xc7z020clg400-1)

Metric	Value	Notes
Errors / warnings / critical warnings	0 / 0 / 0	Clean elaboration
Latches inferred	0	Explicitly checked
Slice LUTs	28	20 logic, 8 distributed RAM
Slice registers	40	All flip-flops
Block RAM / DSP	0 / 0	Memory in distributed RAM

The sixteen-by-eight memory was mapped to distributed RAM rather than a block RAM, which is expected at this size. Vivado's CDC report located both crossings, each at synchroniser depth two and each carrying the `ASYNC_REG` property, confirming that the synchronisers were implemented as intended.

4.3 Hardware Implementation

The design was implemented on a PYNQ-Z2 board. A linear-feedback shift register generates the write data and an identical register on the read side predicts what should be returned, so any mismatch is counted as an error. A second pair of registers stalls both interfaces so that the full and empty conditions are reached, and the counters are read back over JTAG.

The write domain is clocked from `FCLK_CLK0`, produced by the processing system PLL from the 33.333 MHz crystal, and the read domain from the board's separate 125 MHz oscillator. The two sources are independent, so the phase between them drifts continuously rather than repeating, which is the condition the design is intended to tolerate.

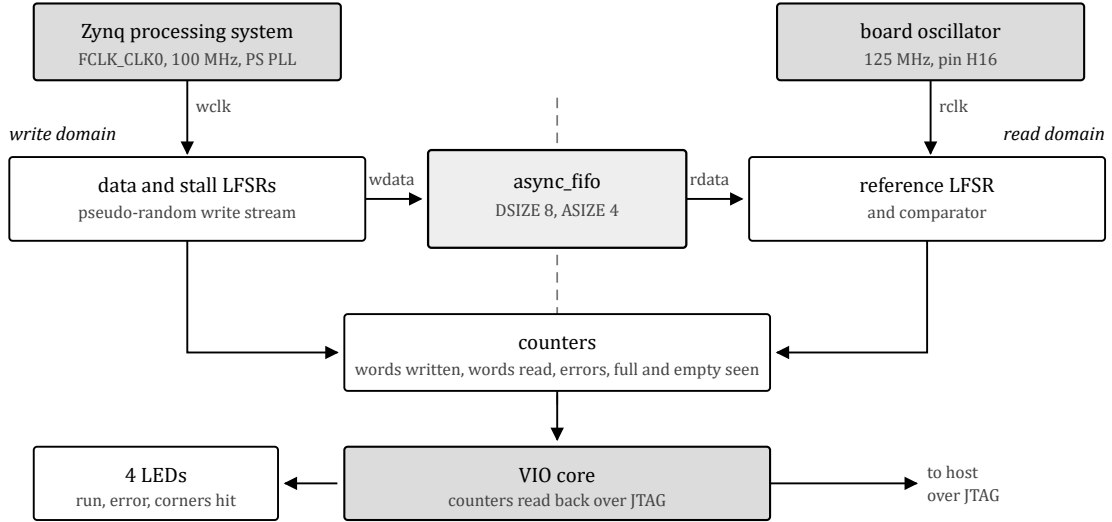


Figure 2: Hardware test setup implemented in the PYNQ-Z2 fabric. The write domain is clocked from the processing system PLL and the read domain from the board oscillator, so the two are independent. A pseudo-random stream is written into the FIFO and an identical reference sequence on the read side is compared against what emerges; any mismatch increments the error counter. Counters are read back over JTAG through a VIO core, and the LEDs give a local indication of the run.

Table 5: Hardware measurement on PYNQ-Z2, two phases of ten seconds

Phase	Words written	Words read	Errors
1 — write fast, read slow	159,905,237	159,905,221	0
2 — write slow, read fast	286,176,459	286,176,459	0
Reached full / empty			yes / yes
Worst setup slack (WNS)			1.997 ns
Worst hold slack (WHS)			0.061 ns

```

-- phase 1  write fast, read slow: w_rate 15/16, r_rate 2/16, 10 s
words written 159905237    read 159905221    errors 0
full seen 1    empty seen 1
-- phase 2  write slow, read fast: w_rate 2/16, r_rate 15/16, 10 s
words written 286176459    read 286176459    errors 0
full seen 1    empty seen 1

=====
HARDWARE RESULT  PYNQ-Z2, xc7z020c1g400-1
=====

wclk  FCLK_CLK0, 100 MHz off the PS PLL
rclk  sys_clk, 125 MHz board oscillator
independent crystals, so the phase between them drifts
words written      286176459
words read+checked 286176459
data errors        0
reached full       yes
reached empty      yes

-----
PASS
=====

```

Figure 3: Console output of the hardware run, captured from the Vivado Tcl session that drove the test over JTAG. The two phases run for ten seconds each and the counters are read back once both interfaces have stopped. The full transcript is in `results/hardware_test.txt`.

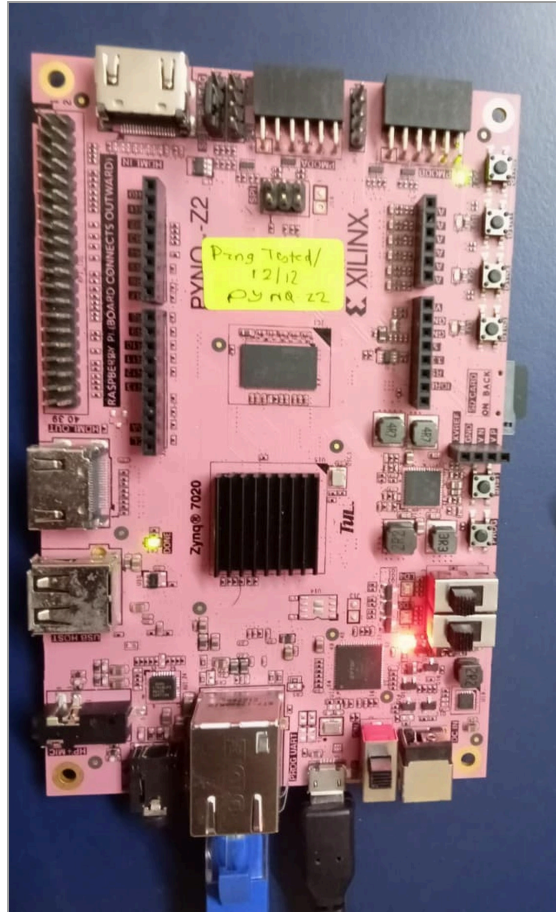


Figure 4: The PYNQ-Z2 with the design loaded and the test running. The green indicator beside the DONE marking shows that the FPGA fabric has been configured, and the user LED near the top of the board is driven by the test logic. The board is attached to the host over the USB JTAG link, which is the only connection the flow uses; no SD card is fitted, because the processing system is started over JTAG solely to supply FCLK_CLK0.

Phase 1 finishes sixteen words short of the number written, which is exactly the depth of the FIFO. That phase ends with the writer running at full rate against a slow reader, so the FIFO is full when writing stops and is still holding sixteen unread words. Phase 2 reverses the rates, the reader drains those words and then keeps pace, and the two totals finish equal.

5. Conclusion

A parameterised dual-clock asynchronous FIFO was designed in Verilog-2001 and verified at three levels. In simulation it transferred 10,017 words with no mismatches across both clock ratios, under two independent simulators that agree on the count, with every coverage point reached. Synthesis for the Zynq XC7Z020 produced no warnings and inferred no latches, and the CDC report confirmed both crossings at synchroniser depth two. On hardware, driven by two independent oscillators, it transferred 286,176,459 words with zero errors while reaching both the full and empty conditions.

The parts of the design requiring most care are the pointer width and the full comparison. The extra pointer bit is what allows full and empty to be distinguished when the addresses are identical, and the full condition requires the top two Gray bits to be inverted rather than one. Deliberately introducing an error into that comparison confirmed that the testbench and the assertions detect it, which is what gives the passing results their value.

References

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- [2] R. Ginosar, “Metastability and synchronizers: A tutorial,” *IEEE Design & Test of Computers*, vol. 28, no. 5, pp. 23–35, Sep./Oct. 2011.
- [3] *IEEE Standard Verilog Hardware Description Language*, IEEE Std 1364-2001, 2001.
- [4] Xilinx Inc., “Zynq-7000 SoC Data Sheet: Overview (DS190),” Xilinx Technical Documentation, 2021.
- [5] AMD/Xilinx, “PYNQ: Python productivity for Zynq — framework documentation,” 2022. [Online]. Available: <http://www.pynq.io>